PCI AI · FORMULA SHEET

# Project Management Formulas

The 16 that separate a plan from a guess

**PERT · CPM · PTA · EMV · WSJF · ROI**

**PROJECT CONTROLS INSTITUTE GLOBAL**

First edition · 2026 · AI proposes. The professional disposes.

PML-AI · HOW TO READ THIS

# Knowing the formula is the easy half

Most project managers can compute a critical path and a CPI.

Far fewer can say when the number stops meaning what they think it means — which is where projects are actually lost.

This deck gives the formula, and the mistake it attracts.

**One rule throughout.** Under hybrid delivery, velocity forecasting and earned value answer the same question through different instruments. State which one governs the commitment before you report either.

ESTIMATING · DOMAIN 6

# Three points, weighted

$$tE = (O + 4M + P) \div 6$$
$$\sigma = (P - O) \div 6$$

PERT weights the most likely case four times.

The triangular alternative, **(O + M + P) ÷ 3**, does not — and gives a different answer. On 8 / 12 / 22, PERT returns **13.0** and triangular returns **14.0**.

**Watch for:** three points supplied by one estimator. That is one number wearing three hats.

CRITICAL PATH · DOMAIN 6

# Float, and a free integrity check

$$EF = ES + \text{Duration}$$
$$LS = LF - \text{Duration}$$
$$TF = LS - ES = LF - EF$$
$$FF = \min(\text{successor } ES) - EF$$

**Total float from the starts must equal total float from the finishes.** If the two forms disagree, your forward or backward pass is wrong. It costs nothing to check and catches a whole class of error.

SCHEDULE COMPRESSION

# Two ways to compress. Different risks.

**Crash cost slope = (Crash cost − Normal cost) ÷ (Normal − Crash duration)**

| | MECHANISM | WHAT IT COSTS YOU |
|---|---|---|
| **Crashing** | Add resource to critical activities | Direct cost, coordination load |
| **Fast-tracking** | Overlap what was sequential | Rework risk rises materially |

**Crashing an activity that is not on the critical path** spends money and saves nothing. It happens more often than anyone admits.

EARNED VALUE · DOMAIN 7

# Where you are, in cost terms

**CPI = EV ÷ AC**
**SPI = EV ÷ PV**
**Cost to date = Actuals + Accruals**

**That third line breaks the first two.** If **AC** is invoiced cost only, work has been performed that nobody has billed you for. **CPI** overstates performance and **EAC** understates cost — and both errors are invisible in the report.

FORECASTING

# Three forecasts. All defensible.

**$EAC = AC + (BAC - EV)$** — the variance was a one-off

**$EAC = BAC \div CPI$** — performance persists

**$EAC = AC + (BAC - EV) \div (CPI \times SPI)$** — cost and schedule both persist

**The arithmetic is trivial. The assumption is the judgement.** State which one you have chosen, publish a range rather than a point, and be ready to defend the choice to a board.

**TCPI**

# The check that exposes wishful budgets

$$\mathrm{TCPI} = (\mathrm{BAC} - \mathrm{EV}) \div (\mathrm{BAC} - \mathrm{AC})$$

The cost efficiency required on every remaining pound to still land on budget.

Now compare it to the **CPI** you have actually achieved.

**Worked:** TCPI 1.08 against a CPI of 0.89. The budget is asserting a 22% efficiency improvement on work that has run 11% below plan. Nobody has demonstrated that. It is a wish with a decimal place.

# Ordinal ratings do not multiply

**Risk score = Probability rating × Impact rating**

A 3 × 4 = 12 heat-map score is a **sorting device**. Nothing more.

It cannot be summed across risks. It cannot size a contingency.

**EMV = Probability × Impact**
**Contingency ≈ Σ EMV, or P80 − P50 from simulation**

**For anything that touches money, use expected value or simulation.** And note that `EMV` describes a portfolio — applied to one binary event, it returns a number that will never occur.

FLOW · DOMAIN 13

# The law that governs every queue

**Cycle time = WIP ÷ Throughput**

12 items in progress, 4 completed per week → **3 weeks** cycle time.

Halve WIP to 6 and cycle time halves to 1.5 weeks, at exactly the same throughput.

**This is why limiting work in progress speeds delivery** without anybody working faster. It is also only valid for a stable system over the window you measured.

VELOCITY

# Forecasting an adaptive delivery

**Velocity = Points completed ÷ Sprint**
**Sprints remaining = Points remaining ÷ Average velocity**
**Capacity = Members × Available days × Focus factor**

**Two failures here.** Using best-ever velocity rather than a rolling average of recent comparable sprints. And setting the focus factor to 1.0, which assumes nobody attends a meeting, answers a question or takes a day off.

REPORTING TO A GATE

# Burn-down hides the thing you need to see

A **burn-down** plots work remaining. Scope added mid-flight looks identical to slow progress.

A **burn-up** plots completed work and total scope as two separate lines.

**Scope growth becomes visible as scope growth.** For any project reporting to a phase gate, use burn-up. The second line is the whole point.

CONTRACTS · DOMAIN 10

# The point where the incentive stops working

**PTA = [(Ceiling price − Target price) ÷ Buyer share ratio] + Target cost**

Above the point of total assumption, the seller absorbs **every further pound**. The buyer's contribution has hit the ceiling.

WORKED · FPIF

# Check your PTA in one line

Target cost 100,000 · target price 113,000 · ceiling 120,000 · buyer share 80%

**PTA** = (7,000 ÷ 0.80) + 100,000 = **108,750**

At an actual cost of exactly 108,750: overrun 8,750, seller absorbs 20% = 1,750, fee = 11,250, final price = **120,000**.

**Exactly the ceiling.** If your PTA does not reconcile to the ceiling price this way, you have used the seller's share ratio in the denominator instead of the buyer's.

COMMUNICATION · DOMAIN 11

# Why small teams move faster

$$\text{Channels} = n(n - 1) \div 2$$

A team of 9 has **36** channels.

Add one person and it becomes **45** — nine new channels for one new person.

**The growth is quadratic, and that is the entire point.** It is the arithmetic behind why adding people to a late project so rarely accelerates it.

QUALITY · DOMAIN 9

# In control and incapable are different states

**Control limits = Mean ± 3σ**
**Process capability = (USL − LSL) ÷ 6σ**

$$\text{Control limits} = \text{Mean} \pm 3\sigma$$

$$\text{Process capability} = (\text{USL} - \text{LSL}) \div 6\sigma$$

**Control limits** describe what your process does.

**Specification limits** describe what the customer requires.

**A process can be perfectly in control and entirely incapable.** The response to each is different, and confusing them sends improvement effort in the wrong direction.

MOST MISAPPLIED · 1–5

# The ten that cost the most

1. Risk scores multiplied and then summed
2. **SV** read as a schedule measure late in a project
3. **PTA** computed with the seller's share ratio
4. Velocity taken as best-ever rather than rolling average
5. Burn-down used for gate reporting

MOST MISAPPLIED · 6–10

# And the other five

1. Standard deviations summed along a path — variances add, not deviations
2. Crashing an activity that is not on the critical path
3. Control limits confused with specification limits
4. Resource utilisation targeted at 100%, leaving nothing to absorb variability
5. Benefits claimed with no pre-delivery baseline ever captured

**Without a baseline, a realisation figure cannot be computed.** It can only be asserted.

# The full sheet is free

This deck is 16 of them. The complete reference carries the whole quantitative surface of project delivery — business case and benefits, estimating, critical path, earned value, risk, agile and flow, procurement and contracts, quality, stakeholders and portfolio — each with the mistake it attracts and reproducible worked examples.

No registration. No email gate.

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